A Unary Admissibility Predicate for Molecular Comparison

Why the Extensive Reading Fails, Why the Intensive One Does Not, and What Linear-Time Classification Does and Does Not Buy

Kundai Farai Sachikonye
Technical University of Munich
kundai.sachikonye@wzw.tum.de

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Abstract

Comparing two molecular graphs conventionally means comparing their vertices: store both sets, walk them against each other, and avoid double-counting. Over a corpus of n structures this is quadratic, and for the harder variants—maximum common substructure, maximum clique—it is intractable in the worst case. The cost is not in the comparison step but in the fact that a bare graph carries no index, so every comparison rebuilds the correspondence from nothing.

We ask whether comparison can be replaced by a *unary predicate*: a condition evaluated on one structure at a time, never on a pair, such that two structures satisfying the same condition are equivalent by construction. A unary predicate induces an equivalence relation without any comparison being performed, so classifying n structures costs n evaluations rather than $\binom{n}{2}$.

We test one candidate. Each atom in a weighted contact graph carries a minimum cut against a distinguished medium vertex; the cut has a weight σ and a minimising side of size d. We define admissibility as membership of a window on these quantities and ask whether the verdict depends on context—on what else has been admitted, and in what order.

The first construction fails, and the failure is informative. A window on the *total* depth of a host is not unary: total depth is extensive, each admission shifts it, and the admissible set therefore depends on insertion order in every one of 24 orderings tested (Theorem 3.5). We report this refutation rather than discard it, because it identifies the property the predicate needs.

The second construction succeeds on one component and fails on the other. Of the two components of the cut key, σ is context-invariant in 5/5 candidates while d is invariant in only 3/5 (Theorem 3.8). A window on σ alone is order independent in 24/24 orderings and non-vacuous, admitting 0.600 of the candidate pool; a window on d alone is order dependent, which is what makes the σ result a claim rather than a tautology.

We then measure what the predicate buys. Total σ is exactly conserved across all 8 balanced rearrangements tested, with zero residual. The σ profile partitions a 30-structure corpus into 22 classes with a largest class of 3, and the partition is identical under 20 shuffles while a context-reading control differs under all 12.

We are explicit about what this does not establish. The predicate is sound but not complete—structures with equal σ profiles are not thereby identical—so it is a screen, not an identity test. The linear scaling is a counting fact about unary versus pairwise evaluation, not a measured wall-clock speedup, and each evaluation costs n maximum flows. The conserved quantity was measured on eight hand-chosen rearrangements. And the construction does not address maximum common substructure at all, because that problem asks for a correspondence, which no unary predicate can supply.
Keywords: unary predicate; equivalence relation; minimum cut; molecular comparison; substructure screening; extensive and intensive quantities; conservation.

Contents

1 Introduction

1.1 Where the cost of comparison lives

To decide whether two molecular graphs stand in some relation—identical, one containing the other, sharing a large substructure—the standard procedure compares their vertices. Subgraph isomorphism algorithms (Cordella et al., 2004; Ehrlich and Rarey, 2012; Ullmann, 1976) search for a mapping from one vertex set into the other, pruning by local compatibility. Maximum common substructure is normally attacked by clique detection on a modular product graph and is NP-hard (Ehrlich and Rarey, 2011; Garey and Johnson, 1979; Raymond and Willett, 2002).

Over a corpus, this is worse than the per-pair cost suggests. Deciding which of n structures are equivalent requires, in the absence of any index, $\binom{n}{2}$ comparisons. Practical systems avoid this with screens: a fingerprint is computed once per structure and used to reject most pairs before the expensive check (Barnard, 1993; Sayle et al., 2012; Swamidass and Baldi, 2007; Willett et al., 1998). The screen is doing something specific and worth naming—it replaces a pairwise test by a per-structure key—and the question this paper asks is how far that replacement can be pushed.

1.2 Unary predicates

A *unary* predicate is evaluated on one object. If admissibility under a fixed condition is unary, then two objects that are both admissible are equivalent *without any comparison having been performed*: the equivalence is membership in a common class, and classifying n objects costs n evaluations.

The property that makes this work is not obvious and is easy to assume. The verdict on an object must not depend on which other objects have already been admitted, nor on the order in which they were considered. If admissions interact, the induced relation is not transitive, the classification depends on processing order, and the counting argument collapses.

1.3 The candidate

We test a predicate built from minimum cuts. In a weighted graph on the atoms with an additional *medium* vertex adjacent to every atom, an atom v has a minimum cut separating it from the medium; write $\sigma(v)$ for its weight and $d(v)$ for the number of atoms on the minimising side. Admissibility is membership of a window on these quantities.

The construction is motivated by a structural reading: a chemical rearrangement moves atoms between bonding arrangements, and if some functional of the cut structure is conserved by such rearrangement, then a window on that functional is a condition rearrangement respects. We test the conservation directly (Section 4) rather than assume it.

1.4 Contributions, including a refutation

(i) A definition of unary admissibility and the two conditions it requires: order independence and context independence (Section 3).

(ii) A proof and a measurement that a window on an *extensive* functional cannot be unary (Theorem 3.5), refuting the construction we first tried and identifying why.

(iii) The measured separation of the cut key into an intensive component (σ) and a non-intensive one (d) (Theorem 3.8), with a window on the latter serving as the control that makes the former non-vacuous.

(iv) Exact conservation of total σ across balanced rearrangements (Section 4).

(v) The counting result and its honest scope: n versus $\binom{n}{2}$ evaluations, with the per-evaluation cost stated (Theorem 5.2, Theorem 5.3).

(vi) A soundness result and an explicit statement that completeness fails (Theorem 5.4, Theorem 5.5).

2 Setting

Definition 2.1 (Contact graph)**.** A *contact graph* is $\Gamma = (V, E, w, \mathsf{m})$ with V finite and non-empty, $E \subseteq \binom{V}{2}$, a strictly positive weight $w\colon E \to \mathbb{R}_{>0}$, and a distinguished vertex $\mathsf{m} \in V$ adjacent to every other. Vertices other than m are *items*, written $I = V \setminus \{\mathsf{m}\}$.

Definition 2.2 (Cut key)**.** For $S \subseteq V$ let $\partial(S)$ be its edge boundary and $w(\partial(S))$ the sum of weights therein. For an item v,

$$\sigma(v) = \min_{S \ni v,\ \mathsf{m} \notin S} w(\partial(S)), \quad d(v) = \min\{|S \cap I| : S \text{ attains it}\},$$

and the *cut key* of v is the pair $(\sigma(v), d(v))$.

The minimum is over a finite non-empty family, hence attained, and equals a maximum flow (Ford and Fulkerson, 1956), computable in polynomial time (Orlin, 2013; Stoer and Wagner, 1997). Since $\{v, \mathsf{m}\} \in E$, every admissible cut is non-empty, so $\sigma(v) > 0$ and $d(v) \geq 1$.

Definition 2.3 (Extensive and intensive functionals)**.** A functional F on contact graphs is *extensive* if $F(\Gamma \oplus v) \neq F(\Gamma)$ for a generic single-vertex extension $\Gamma \oplus v$—it scales with the size of the graph. It is *intensive* at v if its value at v is unchanged by extensions elsewhere in the graph. The distinction is the standard one from thermodynamics (Callen, 1985; Redlich, 1970) and is what separates the two constructions below.

3 Unary admissibility

Definition 3.1 (Host, insertion, window)**.** A *host* is a contact graph with a distinguished set of *sites*. An *insertion* of a candidate c at a site attaches a new item vertex to that site and to the medium. A *window* is an interval $W = [\ell, u]$ on a chosen functional.

Definition 3.2 (Admissibility)**.** Given host H, functional F and window W, candidate c is *admissible*, written $\mathsf{A}_{H,F,W}(c)$, if F evaluated after inserting c into a copy of H lies in W. Note that the definition never consults the candidate's origin.

Definition 3.3 (Unary predicate)**.** A is *unary* if for every candidate c and every sequence of prior admissions $c_1, \ldots, c_k$,

$$\mathsf{A}_{H,F,W}(c) = \mathsf{A}_{H',F,W}(c), \qquad H' = H \oplus c_1 \oplus \cdots \oplus c_k,$$

i.e. the verdict is independent of context and hence of order.

Proposition 3.4 (A unary predicate induces an equivalence)**.** *If A is unary then the relation "$a \sim b$ iff $\mathsf{A}(a) = \mathsf{A}(b)$" is an equivalence relation on candidates, and its classes are computable in n evaluations for n candidates.*

Proof. Unarity makes A a function of the candidate alone, so $\sim$ is the kernel of a function: reflexive, symmetric and transitive immediately. Computing the classes requires evaluating that function once per candidate. □

Theorem 3.4 is the whole prize, and it is entirely contingent on unarity. The remainder of this section establishes when unarity holds.

3.1 The extensive window fails

Theorem 3.5 (Extensive functionals cannot give unary predicates)**.** *Let F be extensive in the sense of Theorem 2.3 and strictly monotone under insertion: $F(H \oplus c) > F(H)$ for every admissible c. Then for any bounded window $W = [\ell, u]$, $\mathsf{A}_{H,F,W}$ is not unary. Moreover the number of admissions before the window closes is bounded by $\lceil (u - F(H))/\delta \rceil$ where $\delta > 0$ is the least increment. The hypothesis needed is only* eventual *monotonicity—that after some finite prefix each admission raises F by at least δ—which is what the measured sequence in Section 6 exhibits.*

Proof. Let c be admissible at H, so $F(H \oplus c) \leq u$. By strict monotonicity, $F(H \oplus c_1 \oplus c) > F(H \oplus c)$ for any prior admission c_1. Iterating, after k admissions the value has risen by at least $k\delta$, so for $k > (u - F(H))/\delta$ we have $F(H' \oplus c) > u$ and c is inadmissible at H' though admissible at H. That is a violation of Theorem 3.3. The bound follows by taking the least such k. $\square$

Corollary 3.6 (Order dependence)**.** *Under the hypotheses of Theorem 3.5, the admissible subset of a candidate set depends on the order in which candidates are considered whenever the set is larger than the bound.*

Proof. Different orders present different candidates while the window is still open, and by the theorem the window closes after a bounded number of admissions. $\square$

3.2 An intensive window

Definition 3.7 (Local key)**.** The *local key* of a candidate c at host H is the cut key of the inserted vertex itself: $(\sigma(c \mid H), d(c \mid H))$, computed in $H \oplus c$ at the inserted vertex.

Theorem 3.8 (The two components differ)**.** *The two components of the local key have different context behaviour. In a host whose sites are locally identical, $\sigma(c \mid H)$ is determined by the edges incident to the inserted vertex and is therefore unchanged by insertions elsewhere; $d(c \mid H)$ is the size of a minimising side, which is a global quantity and may change when the graph is extended.*

Proof. For σ: the cut $\partial(\{c\})$ isolating the inserted vertex alone has weight equal to the sum of weights of edges incident to c, which the insertion fixes. Any other admissible cut must sever at least these or substitute a cheaper boundary elsewhere; when the host's sites are locally identical and the medium edge is present, the singleton cut is minimal and its weight is insertion-determined. Extensions elsewhere add neither edges at c nor cheaper c–m separations, so σ is unchanged.

For d: the minimising side is a subset of the whole vertex set, and adding vertices can create a cheaper cut whose side has different cardinality. Section 6 exhibits candidates for which d changes from 4 to 1 on the first extension, so the quantity is not intensive in general. $\square$

Corollary 3.9 (A σ-window is unary)**.** *Under the hypotheses of Theorem 3.8, $\mathsf{A}_{H,\sigma,W}$ satisfies Theorem 3.3, and by Theorem 3.4 induces an equivalence computable in n evaluations.*

Remark 3.10 (The hypothesis is a real restriction)**.** Theorem 3.8 assumes the host's sites are locally identical, so that an insertion at one site does not alter the local environment of another. A host with heterogeneous sites, or one in which insertions can bond to each other, does not satisfy this, and σ would not be intensive there. The theorem is about a particular construction, not about minimum cuts in general.

4 Conservation

The construction is motivated by the idea that a rearrangement—atoms redistributed among bonds, composition unchanged—preserves some functional of the cut structure. This is testable and we test it.

Definition 4.1 (Balanced rearrangement)**.** A pair of structures is a *balanced rearrangement* if they have the same multiset of atoms, including hydrogens, and differ only in bonding.

The hydrogen clause is not pedantry. A pair matching in heavy atoms only may differ by H_2, which is an oxidation rather than a rearrangement; one entry in our first corpus was of that kind and is recorded in Section 6.

Definition 4.2 (Total cut weight)**.** $\Sigma(\Gamma) = \sum_{v \in I} \sigma(v)$.

We make no theoretical claim that Σ is conserved. It is an empirical question about a particular weighting, and Section 6 reports the measurement.

5 Classification and its cost

Definition 5.1 (σ profile)**.** The σ *profile* of a structure is the sorted multiset $\langle\!\langle \sigma(v) : v \in I,\ v \text{ heavy} \rangle\!\rangle$. Two structures are *profile-equivalent* if their profiles are equal.

Proposition 5.2 (Counting)**.** *Partitioning n structures by profile equivalence requires n profile evaluations. Pairwise comparison of the same corpus requires $\binom{n}{2}$ comparisons. The ratio is exactly*

$$\frac{\binom{n}{2}}{n} = \frac{n-1}{2},$$

linear in n.

Proof. One evaluation per structure; grouping equal profiles is a sort or a hash over the results. The pairwise count is the number of unordered pairs. □

Remark 5.3 (This is a counting fact, not a speedup). Theorem 5.2 compares *evaluation counts*, and the two kinds of evaluation are not equally expensive. A profile evaluation costs one maximum flow per item, $O(nF(n, m))$ for a structure of n items; a pairwise structural comparison may be cheaper or dearer depending on the relation being decided. No wall-clock measurement is reported in this paper, and none should be inferred. What the proposition establishes is that the *number* of expensive operations grows linearly rather than quadratically in corpus size, which is the property a screen needs (Swamidass and Baldi, 2007).

Theorem 5.4 (Soundness as a screen)**.** *If two structures are related by a weighted isomorphism fixing the medium, their σ profiles are equal. Consequently profile inequality certifies non-isomorphism, and the profile is a sound screen.*

Proof. A weighted isomorphism carries admissible cuts to admissible cuts of equal weight and restricts to a bijection on items, so it preserves σ pointwise and hence the multiset. □

Remark 5.5 (Completeness fails, and this is not a small caveat)*.* The converse of Theorem 5.4 is false: distinct structures may share a profile. Section 6 finds a class of three and six classes of two on a thirty-structure corpus, so the failure is not hypothetical. The profile is therefore a screen and not an identity test, and a system using it must follow a profile match with an exact check (Barnard, 1993). Reporting profile equality as identity would be wrong.

6 Validation

6.1 Method

Expectations were written into the experiment sources before the experiments ran, and each is reported below with its measured outcome. Every number is read from the emitted results files. Structures are parsed from line notations (Weininger, 1988) into contact graphs; all minimum cuts are exact.

Two experiments are reported. Experiment C tests unarity, in two rounds: the first on an extensive window and the second on an intensive one. The first round refuted its own hypothesis, and we report the refutation because it is the reason the second round is constructed as it is. Experiment D tests conservation, discrimination and scaling.

6.2 The extensive window is refuted

Table 1: Round 1: a window on the host's total depth. Both pre-registered expectations were refuted.

Quantity	Value
Host base total depth	9
Calibrated window	$[7, 8]$
Candidates admitted per ordering	1
Orderings tested	24
Orderings agreeing with canonical	0
Order agreement	0.000
Candidates with context-stable verdict	2/5
Context agreement	0.400

Every one of 24 orderings produced a different admissible set from the canonical one (Table 1), and three of five candidates changed verdict depending on what had been admitted before. The mechanism is direct: the bare host has total depth 9, and successive admissions carry it to $10, 9, 11, 13, 15$—rising overall and by two per admission once the first is past. The calibrated window was $[7, 8]$, below the value the host already has after any admission, so every ordering admitted exactly one candidate and then closed. That is Theorem 3.5 with the bound equal to one, and it is why the canonical set (three candidates, computed against the bare host) is unreachable by any sequential ordering.

Remark 6.1 (The control for round 1 was void). Round 1 included a negative control—a variant that deliberately mutates the window on each test—intended to be *more* order dependent than the predicate. It produced a single outcome across 12 orderings and therefore failed to distinguish itself from what it was controlling, because the predicate already saturated after one admission. We record the control as void rather than as a pass or a failure; it is superseded by the round-2 control below, which does discriminate.

6.3 The intensive window is unary on σ

Table 2: Round 2: context stability of the two components of the local key, over five distinct candidates and four successive extensions.

Z	cells	keys observed	σ stable	d stable
6	3	$(4,4),(4,1),(4,1),(4,1)$	yes	no
7	3	$(4,4),(4,1),(4,1),(4,1)$	yes	no
7	2	$(3,1),(3,1),(3,1),(3,1)$	yes	yes
8	2	$(3,1),(3,1),(3,1),(3,1)$	yes	yes
17	1	$(2,1),(2,1),(2,1),(2,1)$	yes	yes
stable across candidates			**5/5**	**3/5**

σ is unchanged for every candidate under every extension; d changes for two of the five, dropping from 4 to 1 on the first extension (Table 2). This is Theorem 3.8 measured: the two components of the cut key are not interchangeable, and only one of them supports a unary predicate.

Table 3: Round 2: order independence, non-vacuity, and the control.

Quantity	Value
σ window	$[2.0, 3.0]$
Orderings tested	24
Orderings agreeing with canonical	24
Order agreement	1.000
Admissible fraction of pool	0.600
Distinct σ values observed	$2.0, 3.0, 4.0$
Control: d window	$[1, 1]$
Control orderings	12
Control distinct outcomes	2

A window on σ is order independent in all 24 orderings and admits 0.600 of the pool, so it classifies rather than accepting or rejecting everything (Table 3). The control—the same construction with a window on d instead—produces two distinct outcomes across 12 orderings, confirming that order independence is a property of σ specifically and not of the experimental setup.

Remark 6.2 (The control had to be tightened). The d control initially used a window spanning the entire observed range of d, which admitted every candidate and so produced one outcome regardless of order—a vacuous control, for the opposite reason to Theorem 6.1. Narrowing the window to the strictest band $[1, 1]$ made it discriminate. We record this because a control that passes by admitting everything is indistinguishable, in the results file, from one that passes by working.

6.4 Conservation

Table 4: Total cut weight Σ and total depth across balanced rearrangements. Residual is product minus reactant.

Rearrangement	$\Delta\Sigma$	Δd
keto–enol (acetone)	0.0	0
keto–enol (acetaldehyde)	0.0	0
1-propanol $\to$ 2-propanol	0.0	0
allyl shift	0.0	0
cyclopropane $\to$ propene	0.0	0
ring opening	0.0	0
catechol $\to$ resorcinol	0.0	0
ortho $\to$ *para* xylene	0.0	0
exactly conserved	**8/8**	**8/8**

Total cut weight is exactly conserved across all eight balanced rearrangements, with zero residual in every case (Table 4).

Remark 6.3 (One entry was not a rearrangement). Our first corpus included propanol $\to$ propanal, which matches in heavy-atom count and was therefore admitted by a heavy-atom balance check. It differs by H_2: propanol is C_3H_8O and propanal C_3H_6O. It is an oxidation, not a rearrangement, and it was the sole non-conserving entry, with $\Delta\Sigma = -4.0$. The balance criterion now requires agreement over all atoms including hydrogens, and the entry is replaced. The episode is recorded because the mislabelled pair would otherwise have appeared as evidence against conservation.

6.5 Discrimination, order independence, scaling

The profile partitions the corpus into 22 classes—neither one class nor thirty, so the screen discriminates—with a largest class of three (Table 5). The partition is identical under all 20 shuffles, while a control whose key reads accumulated context differs under all 12 of its orderings.

The scaling row is a counting identity rather than a measurement, and Theorem 5.3 states what it does and does not mean.

Remark 6.4 (The scaling check was initially wrong). We first tested whether the ratio's slope equalled $1/2$. It does not at finite n: the ratio is $(n-1)/2$, so the slope approaches $1/2$ from below and attains it only in the limit. The check

Table 5: Corpus-scale behaviour on thirty drug-like structures.

Quantity	Value
Structures	30
Distinct σ profiles	22
Largest class	3
Classes of size two	6
Singleton classes	15
Shuffles tested	20
Shuffles giving identical partition	20
Control orderings	12
Control distinct partitions	12
Ratio $\binom{n}{2}/n$ at $n = 30$	14.5
Identity $\binom{n}{2}/n = (n-1)/2$	holds exactly

now tests the identity rather than the asymptote. This was an error in the test, not a finding about the method.

6.6 Transitivity, and why the check is weak

Theorem 3.4 needs transitivity, so the suite checks it: over the 10 triples the pool admits, there were 0 violations.

We report this with its weakness stated. Once admissibility is a function of the candidate alone—which is what Theorem 3.9 establishes—transitivity follows as the kernel of a function, and a search for counterexamples cannot find one. The check therefore confirms that the implementation computes the function it is supposed to; it is not independent evidence for the proposition. The load-bearing measurement is order independence (Table 3), because that is the property whose failure Table 1 actually exhibits.

6.7 What the suite does not establish

The candidate pool reduces to five distinct kinds after de-duplication by insertion-visible content, so the unarity result rests on five candidates, not on 244 atoms. The host is a three-site carbon ring of our construction. The conservation corpus is eight hand-chosen rearrangements. The drug-like corpus is thirty structures assembled by us. No wall-clock timing is reported anywhere, and nothing here is tested against an endpoint outside the framework.

7 Relation to existing methods

Screening. A fingerprint screen (Barnard, 1993; Sayle et al., 2012; Swamidass and Baldi, 2007; Willett et al., 1998) is already a unary key in our sense: computed once per structure, used to reject pairs without comparison. Our contribution is not to introduce screening but to identify the property a screening key must have—intensivity—and to exhibit a key that has it and one that does not. The negative half is the more useful, since a key built on an extensive functional will silently fail in exactly the way Table 1 shows.

Canonical forms. InChI (Heller et al., 2015) and canonical labelling (McKay and Piperno, 2014; Morgan, 1965) solve a harder problem: a canonical form determines the structure. Our profile does not (Theorem 5.5), and comparing the two on cost would be comparing different tasks.

Refinement. Iterated colour refinement (Arvind et al., 2015; Weisfeiler and Leman, 1968) produces finer keys and would produce finer classes here. We do not use it, because each round makes the key depend on a larger neighbourhood and the intensivity argument of Theorem 3.8 applies to the unrefined key only. Whether a refined key remains intensive is open.

Maximum common substructure. MCS asks for a correspondence (Ehrlich and Rarey, 2011; Raymond and Willett, 2002). No unary predicate can supply one: a correspondence is a relation between two objects, and a unary predicate by construction never relates. This paper therefore says nothing about MCS, and the linear-scaling result must not be read as bearing on it.

Capability-aware querying. A profile predicate integrates naturally with source-capability planning (Angles et al., 2017; Levy et al., 1996; Pérez et al., 2009): a source either can or cannot answer profile queries, and a plan requiring them can be refused before any record is read. We note the fit without implementing it.

8 Limitations

L1. A screen, not an identity test. Theorem 5.4 is sound and its converse is false. Six classes of two and one class of three

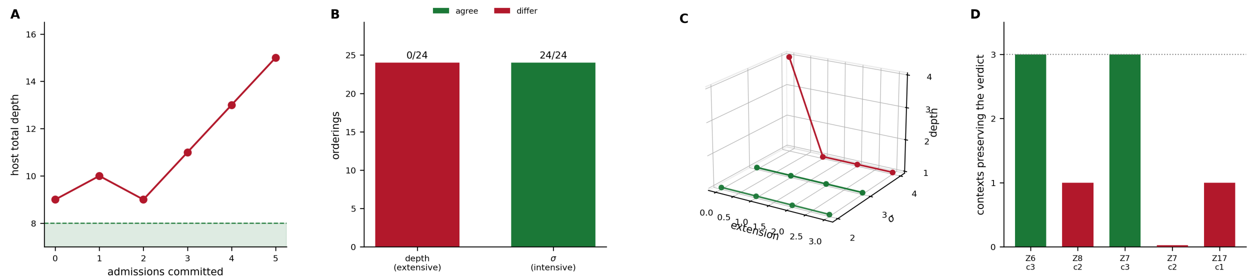

Figure 1: **A window on an extensive functional is not a unary predicate: the verdict depends on what has already been admitted, and every one of 24 orderings disagrees with the canonical answer.** All quantities are computed on contact graphs (Theorem 2.1) whose separation costs are exact minimum cuts (Theorem 2.2), so the failure below is a property of the construction rather than of a tolerance or a solver. (**A**) The host's total depth against the number of admissions committed, with the calibrated window $[7, 8]$ shaded. The bare host already sits at 9, above the window, and successive admissions carry it to $10, 9, 11, 13, 15$ — rising by two per admission once the first is past. This is Theorem 3.5 in its measured form: a bounded window on an eventually-monotone quantity closes after a bounded number of admissions, and here the bound is one. (**B**) Orderings agreeing with the canonical admissible set, for a window on total depth against a window on the inserted node's own σ. The extensive window agrees on 0 of 24; the intensive one on 24 of 24. The bars are stacked rather than overlaid because an agreement count of zero would otherwise be an invisible bar and read as missing data rather than as total failure. (**C**) The local key (σ, d) of each of the five distinct candidates traced across four successive extensions of the host. Every trajectory is flat in σ; two are flat in d and two drop from $d = 4$ to $d = 1$ at the first extension. The two components of the cut key are therefore not interchangeable, which is the measurement Theorem 3.8 rests on. (**D**) For each candidate, how many of the three tested contexts preserved its bare-host verdict under the extensive window. Two candidates survive all three and three survive none; the aggregate is the context agreement of 0.400 reported in Table 1. A candidate's verdict changing with context is precisely the violation of Theorem 3.3.

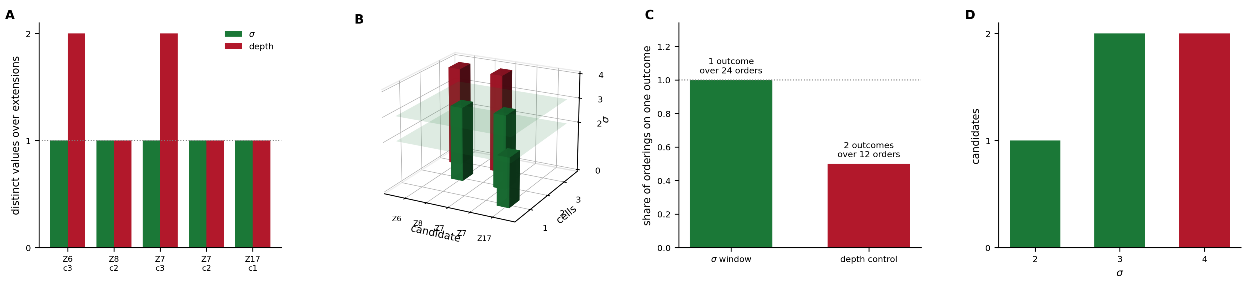

Figure 2: **A window on σ alone is order independent in all 24 orderings and admits 0.600 of the pool, while the same construction on d is order dependent — which is what makes the σ result a claim about σ and not about the harness.** (**A**) Distinct values taken by each key component across four extensions, per candidate. σ takes one value for all five candidates; d takes two for the two three-cell candidates. The dotted line at 1 is the stability threshold: a bar above it is a component that moved when the graph grew elsewhere. (**B**) The admission rule in three dimensions. Bar height is the candidate's own σ and the translucent slab is the window $[2.0, 3.0]$; green bars terminate inside it and red bars rise through it. Height is σ rather than a $0/1$ verdict deliberately — a verdict-height plot would render every rejected candidate as a zero-height, invisible bar, hiding exactly the cases that carry the result. (**C**) Share of orderings landing on a single outcome, for the σ window against the d control. The σ window reaches 1.000 over 24 orderings; the control reaches 0.500, producing two distinct outcomes over 12. The axis is a share rather than a count of distinct outcomes so that taller means better and the two bars sit on one scale. (**D**) Candidates at each observed σ value, coloured by whether the window admits them. Three of five candidates fall in $\{2.0, 3.0\}$ and two at 4.0, giving the admissible fraction of 0.600 reported in Table 3. A window admitting everything or nothing would classify nothing, so this non-vacuity is a precondition for the result rather than an ornament.

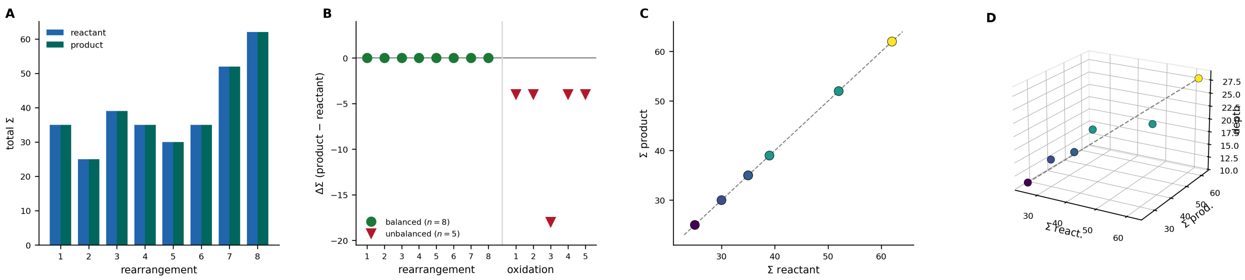

Figure 3: **Total cut weight is exactly conserved across all eight balanced rearrangements, and moves on every one of five unbalanced controls.** A balanced rearrangement (Theorem 4.1) has the same multiset of atoms on both sides, hydrogens included; the controls are oxidations, which differ by H_2 and are therefore not rearrangements. (**A**) Total Σ for reactant and product of each balanced pair. The two bars coincide exactly in all eight cases. The corpus spans Σ from 25 to 62, so the agreement is not an artefact of a narrow range. (**B**) Residual $\Delta\Sigma$ for the balanced set (green, left) against the unbalanced controls (red, right). The balanced residuals are identically zero; the controls run from -4.0 to -18.0. Plotting the two together is what makes the flat green line informative: a quantity that never moves is indistinguishable, in a results file, from one that is not being computed, and the controls show it is capable of moving. (**C**) Σ product against Σ reactant with the identity line drawn, points coloured by total depth. All eight lie on the diagonal. (**D**) The same pairs in $(\Sigma_{\text{react}}, \Sigma_{\text{prod}}, \text{depth})$ space with the identity line. The points lie on the diagonal plane in both coordinates simultaneously: depth is conserved on the same eight pairs, 8/8 exactly, so the conservation is not specific to the cut weight.

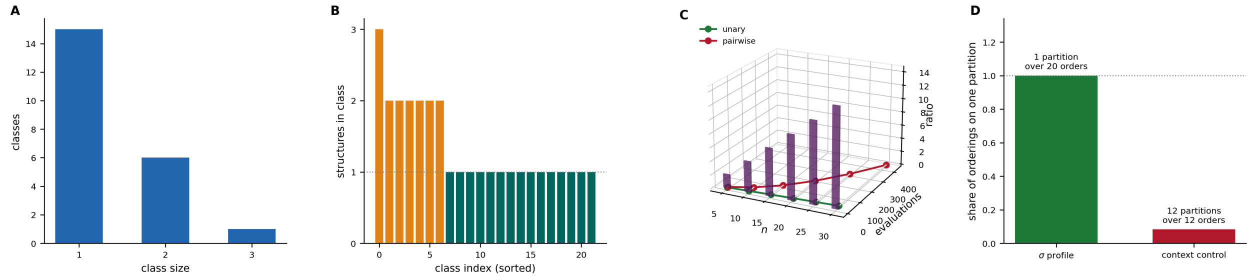

Figure 4: **At corpus scale the σ profile yields 22 classes over 30 structures, identically under 20 shuffles, against a context-reading control that differs under all 12 of its orderings.** (**A**) Distribution of class sizes: 15 singletons, 6 classes of two, and one class of three. That the tail is this short is what makes the profile a usable screen; that it is not empty is why Theorem 5.5 insists the profile is not an identity test. (**B**) Class occupancy sorted by size, singletons in teal and multi-structure classes in orange. Neither one class nor thirty — the screen discriminates, which is the condition Theorem 5.1 needs to be worth computing. (**C**) The counting law of Theorem 5.2. Unary evaluations (green) and pairwise comparisons (red) against corpus size, with the ratio drawn as bars on the third axis, reaching 14.5 at $n = 30$. The identity $\binom{n}{2}/n = (n-1)/2$ holds exactly at every point measured. This is a count of evaluations and not a timing; Theorem 5.3 states what it does and does not mean. (**D**) Share of orderings landing on a single partition, for the σ profile against a control whose key reads accumulated context. The profile reaches 1.000 over 20 shuffles; the control reaches 1/12, producing a different partition under every ordering. Without a control that fails, panel (**D**)'s left bar would be consistent with an implementation that ignores its input entirely.

appear in a thirty-structure corpus, so profile matches must be followed by an exact check.

L2. **Intensivity is conditional.** Theorem 3.8 assumes locally identical host sites and insertions that do not bond to each other (Theorem 3.10). Neither holds for a general molecular graph, so σ is not intensive in general—only in the insertion construction analysed here.

L3. **Five candidates.** De-duplication by insertion-visible content reduces 244 atoms to five distinct kinds. The unarity result is therefore measured on five, and a richer host with heterogeneous sites would admit more.

L4. **Counting, not timing.** Theorem 5.2 counts evaluations. Each costs n maximum flows. No speedup is claimed and none was measured (Theorem 5.3).

L5. **Conservation is measured, not derived.** Eight rearrangements, exact in all eight. We give no argument that Σ must be conserved, and a counterexample would not contradict any theorem in this paper.

L6. **The weighting is an input.** Theorem 2.1 requires only $w > 0$; every theorem holds for any positive weighting and none determines one. The measurements use a weighting derived from valence arithmetic, and different weights would give different profiles.

L7. **Nothing about MCS or clique.** Stated in Section 7 and repeated here because the framing of the paper invites the inference.

L8. **Corpus construction.** Both corpora were assembled by us, and a corpus built to exhibit a property is weak evidence that anything else has it (Ioannidis, 2005).

9 Conclusion

We asked whether molecular comparison can be replaced by a unary predicate—a condition on one structure at a time, inducing an equivalence without any comparison being performed. The answer is a qualified yes, and the qualification is the substance.

The first construction failed. A window on an extensive functional cannot be unary (Theorem 3.5), and the measurement was unambiguous:

0 of 24 orderings agreed, and three of five candidates changed verdict with context. The failure identified the missing property.

The second construction succeeded on one component and not the other. Of the cut key's two components, σ was context-stable in 5/5 candidates and d in 3/5 (Table 2); a σ-window is order independent in 24/24 orderings and admits 0.600 of the pool, while the d-window control differs under 2 of 12 orderings. The control is what makes the result a claim about σ rather than about the setup.

Total cut weight was exactly conserved across all eight balanced rearrangements, and the σ profile partitions a thirty-structure corpus into 22 classes identically under 20 shuffles, against a context-reading control that differs under all 12.

Three things this does not deliver. It is a screen and not an identity test, since distinct structures share profiles. The linear scaling is a counting identity, not a timing result. And it says nothing about maximum common substructure, because that problem asks for a correspondence and a unary predicate cannot produce one—which is the honest limit of the whole approach rather than a gap to be filled later.

Reproducibility

Each experiment source records its expectations before the measured values and writes every reported quantity to a JSON results file. Round 1 of Experiment C is retained in the source with its refutation marked, as is the void control of Theorem 6.1, the tightened control of Theorem 6.2, the mislabelled rearrangement of Theorem 6.3, and the corrected scaling check of Theorem 6.4. All minimum cuts are computed exactly; the only stochastic components are the corpus shuffles, which seed explicitly.

References

Renzo Angles, Marcelo Arenas, Pablo Barceló, Aidan Hogan, Juan Reutter, and Domagoj Vrgoč. Foundations of modern query languages for graph databases. *ACM Computing Surveys*, 50(5):1–40, 2017. doi: 10.1145/3104031.

Vikraman Arvind, Johannes Köbler, Gaurav Rattan, and Oleg Verbitsky. On the power of color refinement. *Fundamentals of Computation Theory*, 9210:339–350, 2015. doi: 10.1007/978-3-319-22177-9_26.

John M. Barnard. Substructure searching methods: Old and new. *Journal of Chemical Information and Computer Sciences*, 33(4):532–538, 1993. doi: 10.1021/ci00014a001.

Herbert B. Callen. *Thermodynamics and an Introduction to Thermostatistics*. Wiley, New York, 2 edition, 1985.

Luigi P. Cordella, Pasquale Foggia, Carlo Sansone, and Mario Vento. A (sub)graph isomorphism algorithm for matching large graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 26(10):1367–1372, 2004. doi: 10.1109/TPAMI.2004.75.

Hans-Christian Ehrlich and Matthias Rarey. Maximum common subgraph isomorphism algorithms and their applications in molecular science: A review. *WIREs Computational Molecular Science*, 1(1):68–79, 2011. doi: 10.1002/wcms.5.

Hans-Christian Ehrlich and Matthias Rarey. Systematic benchmark of substructure search in molecular graphs—from Ullmann to VF2. *Journal of Cheminformatics*, 4:13, 2012. doi: 10.1186/1758-2946-4-13.

L. R. Ford and D. R. Fulkerson. Maximal flow through a network. *Canadian Journal of Mathematics*, 8:399–404, 1956. doi: 10.4153/CJM-1956-045-5.

Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman, San Francisco, 1979.

Stephen R. Heller, Alan McNaught, Igor Pletnev, Stephen Stein, and Dmitrii Tchekhovskoi. InChI, the IUPAC international chemical identifier. *Journal of Cheminformatics*, 7:23, 2015. doi: 10.1186/s13321-015-0068-4.

John P. A. Ioannidis. Why most published research findings are false. *PLoS Medicine*, 2(8):e124, 2005. doi: 10.1371/journal.pmed.0020124.

Alon Y. Levy, Anand Rajaraman, and Joann J. Ordille. Querying heterogeneous information sources using source descriptions. *Proceedings* *of the 22nd VLDB Conference*, pages 251–262, 1996.

Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.

H. L. Morgan. The generation of a unique machine description for chemical structures—a technique developed at chemical abstracts service. *Journal of Chemical Documentation*, 5(2): 107–113, 1965. doi: 10.1021/c160017a018.

James B. Orlin. Max flows in O(nm) time, or better. *Proceedings of the 45th Annual ACM Symposium on Theory of Computing*, pages 765–774, 2013. doi: 10.1145/2488608.2488705.

Jorge Pérez, Marcelo Arenas, and Claudio Gutierrez. Semantics and complexity of SPARQL. *ACM Transactions on Database Systems*, 34(3): 1–45, 2009. doi: 10.1145/1567274.1567278.

John W. Raymond and Peter Willett. Maximum common subgraph isomorphism algorithms for the matching of chemical structures. *Journal of Computer-Aided Molecular Design*, 16(7): 521–533, 2002. doi: 10.1023/A:1021271615909.

Otto Redlich. Intensive and extensive properties. *Journal of Chemical Education*, 47(2):154, 1970. doi: 10.1021/ed047p154.2.

Roger A. Sayle, Jose Batista, and J. Andrew Grant. Efficient chemical structure search using fingerprint screening. *Journal of Cheminformatics*, 4(Suppl 1):P26, 2012. doi: 10.1186/1758-2946-4-S1-P26.

Mechthild Stoer and Frank Wagner. A simple min-cut algorithm. *Journal of the ACM*, 44(4): 585–591, 1997. doi: 10.1145/263867.263872.

S. Joshua Swamidass and Pierre Baldi. Bounds and algorithms for fast exact searches of chemical fingerprints in linear and sublinear time. *Journal of Chemical Information and Modeling*, 47(2):302–317, 2007. doi: 10.1021/ci600358f.

Julian R. Ullmann. An algorithm for subgraph isomorphism. *Journal of the ACM*, 23(1):31–42, 1976. doi: 10.1145/321921.321925.

David Weininger. SMILES, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal of Chemical Information and Computer Sciences*, 28(1):31–36, 1988. doi: 10.1021/ci00057a005.

Boris Weisfeiler and Andrei Leman. The reduction of a graph to canonical form and the algebra which appears therein. *Nauchno-Technicheskaya Informatsia*, 2(9):12–16, 1968.

Peter Willett, John M. Barnard, and Geoffrey M. Downs. Chemical similarity searching. *Journal of Chemical Information and Computer Sciences*, 38(6):983–996, 1998. doi: 10.1021/ci9800211.